

ELI — Architecture Ascii

ELI v2.3.20

August 2026

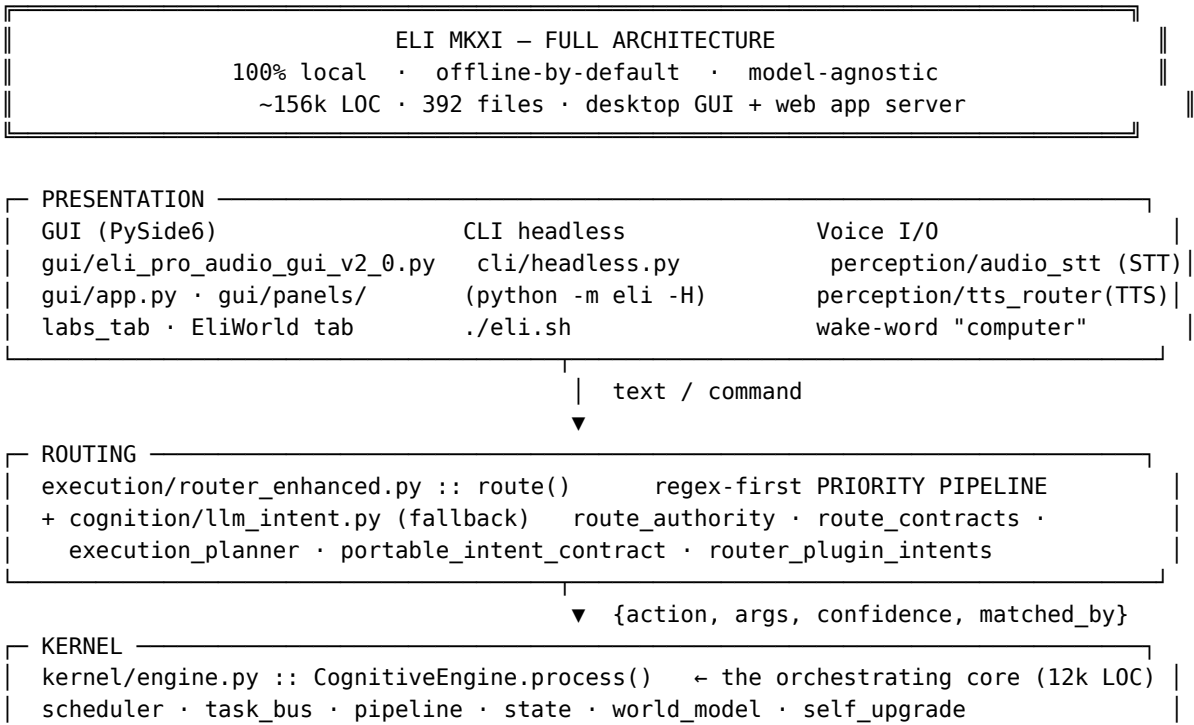
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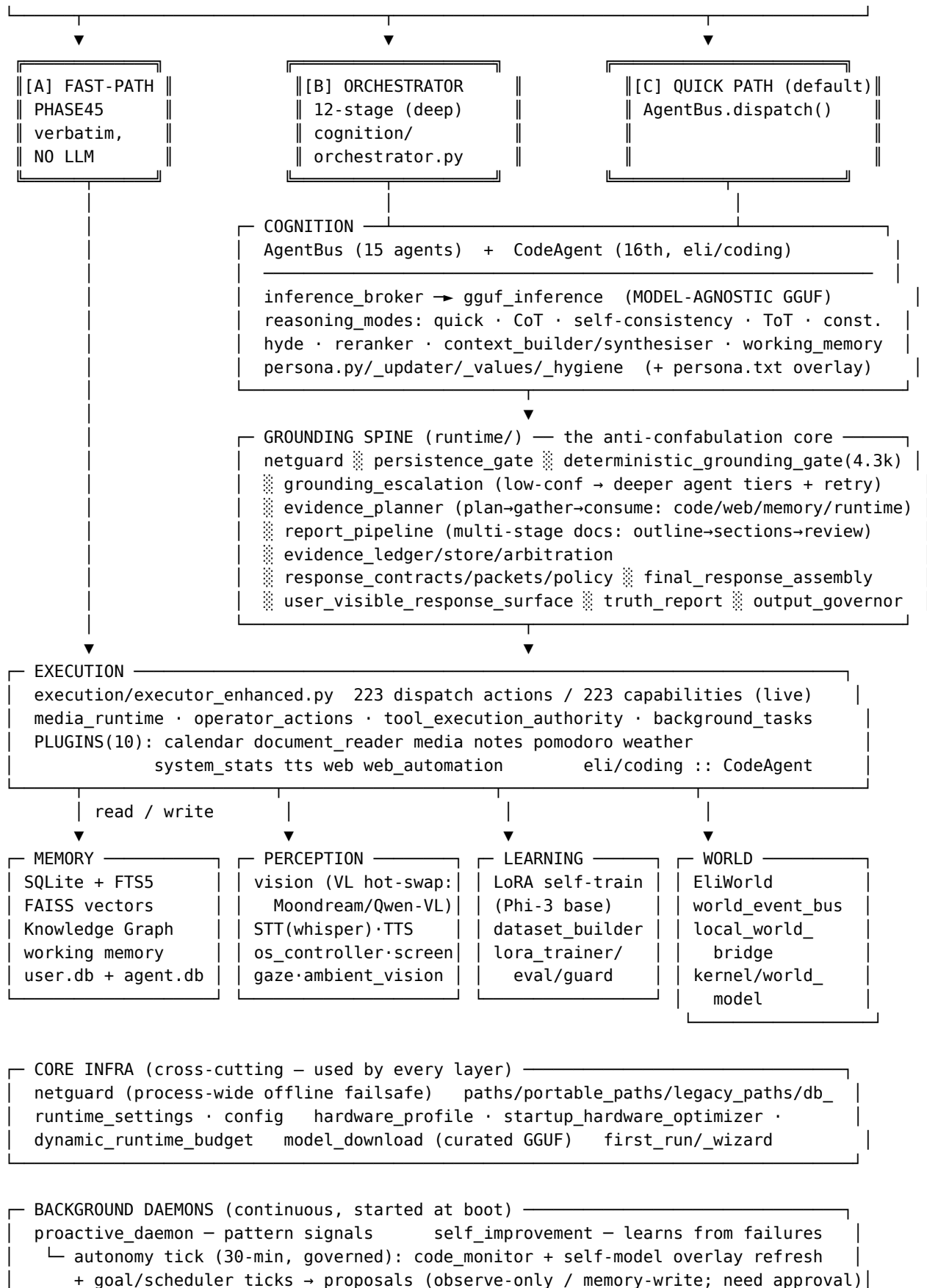
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Blueprint — ELI MKXI Full Architecture (ASCII)

The entire system in one drawing, plus the module tree and data layout. Grounded in the real source (see architecture.md for prose, diagrams.md for the pipeline/memory/gating close-ups). Every layer and box maps to a real path.

A. Full system — layered block diagram





habits_scheduler/habits – routines	scheduler · task_bus – jobs
background_tasks – async heavy work	reflection loop ambient_vision loop
scheduled_tasks (durable overnight/timed)	world_event_bus ← confidence events

```

OUTPUT ← output_governor / sanitiser ← inference ← (any path)
      |
      └─ TTS (Piper) / GUI render

```

B. The 15 agents

AGENT BUS (cognition/agent_bus.py :: AgentBus.dispatch → DispatchResult)
 selects a set → runs on a dependency DAG → aggregates grounding_confidence

BusMemoryAgent	recall: SQLite/FTS5/FAISS → context
KnowledgeGraphAgent	entities/relations from the KG
SystemAgent	runs executor actions (WEB_SEARCH, RUNTIME_AUDIT, ...)
OrchestratorAgent	plan / coordinate grounded synthesis
FileCodeAgent	searches the codebase for code-grounded answers
IntrospectionBus...	live runtime + gathers identity/awareness audits as evidence (persona summarises – never a data dump)
CapabilityAgent	capability manifest (what ELI can do)
ReflectionAgent	session reflection / insights
ProactiveAgent	proactive patterns / signals
HabitAgent	learned habits
SelfImprovementAgent	self-improvement state
FrontierAgent	frontier reasoning hooks
PluginAgent	dispatch to registered plugins
VoiceAgent	STT/TTS engine state
CriticAgent	second-tier verifier: checks retriever agents' output
+ CodeAgent (16th)	SEPARATE pipeline: plan→search→verify→repair
eli/coding/agent	(CODE_SOLVE / GENERATE_SCRIPT) – not a bus agent

C. Module tree (LOC · key files · role)

```

eli/ (~156k LOC, 392 files) · api/server.py (FastAPI web app + dashboard)
├── __main__.py ..... entry dispatch (GUI | --headless)
├── kernel/          13.1k — the core
│   ├── engine.py    12.0k   CognitiveEngine.process() – the spine *god-file
│   ├── scheduler.py          timed jobs
│   └── task_bus.py · pipeline.py · state.py · world_model.py · self_upgrade.py
├── execution/       20.5k — route → act
│   ├── executor_enhanced.py  15.0k  223 actions *god-file
│   ├── router_enhanced.py    6.1k   priority pipeline *god-file
│   ├── execution_planner.py · route_authority.py · route_contracts.py
│   ├── tool_execution_authority.py · operator_actions.py · operator_policy.py
│   └── media_runtime.py · router_plugin_intents.py · executor_plugin_handlers.py

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- └─ portable_intent_contract.py · execution_intent_packets.py
- cognition/ 12.3k — think
 - └─ agent_bus.py 2.4k 15 agents + dispatch *
 - └─ orchestrator.py 12-stage deep retrieval
 - └─ gguf_inference.py 2.1k · inference_broker.py model-agnostic inference
 - └─ reasoning_modes.py · hyde.py · reranker.py · llm_intent.py
 - └─ persona.py/_updater/_values/_status/_hygiene (+ persona.txt, persona.auto.txt)
 - └─ context_builder.py · context_synthesiser.py · working_memory.py
 - └─ output_governor.py · response_governance.py · response_sanitizer.py
 - └─ grounded_status.py · introspection_agent.py · tone_analyzer.py · chat_model.py
- runtime/ 18.0k — grounding spine + daemons (68 files)
 - └─ deterministic_grounding_gate.py 4.3k * · grounding_escalation.py
 - └─ evidence_ledger/store/arbitration.py · memory_evidence.py
 - └─ persistence_gate.py · truth_report.py · control_contracts.py
 - └─ response_contracts/packets/policy.py · final_response_assembly/provider.py
 - └─ user_visible_response_surface.py · personal_memory_*.py · reflection.py
 - └─ background_tasks.py · self_improvement.py · code_monitor.py
 - └─ capability_sync.py · pending_proposal.py · runtime_policy.py
- memory/ 6.6k — remember (13 files)
 - └─ memory.py 4.5k Memory · SQLite + FTS5 *god-file
 - └─ vector_store.py FAISS index
- perception/ 5.4k — sense (18 files)
 - └─ vision.py · analyze_image/csv/pdfs/mesh.py · ambient_vision.py
 - └─ audio_stt.py · local_whisper_stt.py · voice_worker(_streaming).py · eli_listen.py
 - └─ tts_router.py · os_controller.py · screen_locator.py · gaze_engine.py
- planning/ 3.1k — proactivity (21 files)
 - └─ proactive_daemon.py · habits_scheduler.py · habits.py · jobqueue_cli.py
- coding/ 1.5k — CodeAgent (plan→search→verify→repair, 9 files)
- learning/ 3.1k — LoRA self-training (11 files)
 - └─ lora_trainer/eval/guard.py · dataset_builder/filters.py
 - └─ bootstrap_phi3_base.py · base_model_resolver.py · training_preflight.py
- plugins/ 1.9k — manager + 10 plugins
 - └─ calendar · document_reader · media · notes · pomodoro · weather · system_stats · tts · web · web_automation
- world/ 1.5k — EliWorld (world_event_bus, local_world_bridge)
- core/ 4.9k — infra
 - └─ netguard.py offline failsafe + allow_network()
 - └─ paths.py · portable_paths.py · legacy_paths.py · db_paths.py
 - └─ runtime_settings.py · config.py · grounding.py
 - └─ hardware_profile.py · startup_hardware_optimizer.py · dynamic_runtime_budget.py
 - └─ model_download.py · first_run.py · first_run_wizard.py
- gui/ 18.7k — PySide6 desktop
 - └─ eli_pro_audio_gui_v2_0.py 11.0k *god-file · app.py · labs_tab.py 5.7k
 - └─ panels/ (startup.py: model picker + FirstBootWizard, HardwareTuningDock)

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├─ tools/          8.1k — image_engine · news · document tools
├─ contracts/ 0.7k · cli/ 0.1k · system/ 0.3k · utils/ 0.9k

```

D. On-disk data layout

```

artifacts/
├─ db/
│   ├── user.sqlite3      memories(+_fts) · conversation_turns · conversations ·
│   │                     kg_entities(+_fts) · kg_relations · recall_log ·
│   │                     runtime_events · news_articles(+_fts) · news_reflections ·
│   │                     habits/habit_events/habit_rules · observations ·
│   │                     learning_replay · working_memory_pins · user_patterns ·
│   │                     session_summaries · corrections · failures
│   └─ agent.sqlite3      agent_dispatches · agent_metrics · improvements ·
│                           failures · code_patches · error_tracking · observations
├─ vectors/index.faiss    semantic index (embedder: nomic-embed-...Q4_K_M.gguf)
├─ conversations/ + archive/
├─ runtime/users/<uuid>/user_profile...    per-user profile
├─ runtime_snapshot.json    live model/runtime truth
├─ world/{snapshots,ledger,journal}/ · proactive/ · incidents/<date>.jsonl
├─ documents/ · scripts/ · analyze_image_*/
|
config/  settings.json (gitignored) · templates/settings.template.json ·
         settings.example.json · plugins_state.json
models/  <your>.gguf · embeddings/ · whisper/ · image/          voices/ *.onnx (Piper)

```

E. The flow in one line

```

INPUT → ROUTER → ENGINE ─┬─ FAST-PATH ───────────────────→ OUTPUT (verbatim)
                        │
                        │ ┌─ ORCHESTRATOR → retrieval ─┐
                        │ │ └─ QUICK → AGENT BUS ───┐
                        │ └────────────────────────┘
MEMORY (SQL/FTS5/FAISS/KG/WM) → context ←──────────┘
GATES: netguard · persistence · grounding · escalation · confidence · governor
INFERENCE (broker → gguf, model-agnostic) → OUTPUT → TTS/GUI
BACKGROUND: proactive · self-improve · habits · learning(LoRA) · world · scheduler

Companion docs: architecture.md (prose, every subsystem) · diagrams.md (pipeline / memory
/ gating close-ups) · tools/eval/README.md (the measurement layer). “”

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